

Clinical Risk Assessment Report: Patient 66

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 34.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.57x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +6.7%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT)	Value: -0.4 [Bottom 3%] (+6.7%)
<i>Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.</i>	
Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET)	Value: 0.72 [Top 13%] (+5.8%)
<i>Points to continuous, elongated bands of high-density or highly active tumor tissue.</i>	
Small Zone Emphasis (PET) (GLZLM_SZE.PET)	Value: 1.27 [Top 10%] (+3.2%)
<i>Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.</i>	
75th Percentile Tumor Density (CONVENTIONAL_HUQ3)	Value: 0.63 [Avg (66th %ile)] (+2.5%)
<i>Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.</i>	
Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT)	Value: -0.25 [Bottom 4%] (+2.2%)
<i>Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.</i>	
High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT)	Value: 0.87 [Top 20%] (+1.7%)
<i>Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.</i>	

3. Protective / Mitigating Factors (Reducing Risk)

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT)	Value: 1.03 [Top 15%] (-3.6%)
<i>Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.</i>	
Squamous Cell Carcinoma Antigen (SCC_Ag)	Value: 0.9 [Avg (74th %ile)] (-1.8%)
<i>Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.</i>	

Machine Learning Feature Attribution (SHAP Decision Path)

